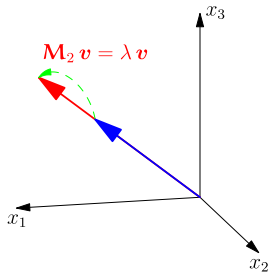


Eigensystems

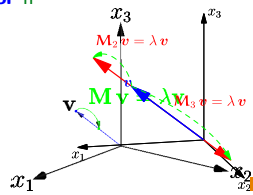
Eigenvectors, Orthogonal Matrices, Eigenvector Decomposition, Rank

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Eigenvector equation

- Eigen-systems help us to understand mappings
- A vector v is said to be an **eigenvector** if

$$Mv = \lambda v$$



- M is square (i.e. $n \times n$)
- Where the number λ is the **eigenvalue**
- Eigenvalues play a fundamental role in understanding operators

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Proof of Orthogonality

- $(Mv_i = \lambda_i v_i)^T$ implies $v_i^T M^T = \lambda_i v_i^T$
- When M is symmetric then $Mv_i = \lambda_i v_i \Rightarrow v_i^T M = \lambda_i v_i^T$
- Consider two eigenvectors v_i and v_j of M

$$\begin{aligned} v_i^T M v_j &= (v_i^T M) v_j = \lambda_i v_i^T v_j \\ &= v_i^T (M v_j) = \lambda_j v_i^T v_j \end{aligned}$$

- So either $\lambda_i = \lambda_j$ or $v_i^T v_j = 0$
- If $\lambda_i = \lambda_j$ then any linear combination of v_i and v_j is an eigenvector ($M(av_i + bv_j) = \lambda_i(av_i + bv_j)$) So I can choose two eigenvectors that are orthogonal to each other.

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Orthogonal Matrices

- We can construct an **orthogonal** matrix V from the eigenvectors

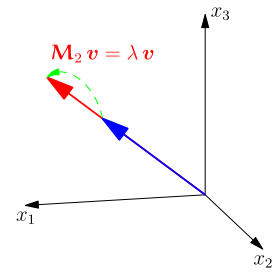
$$V = (v_1, v_2, \dots, v_n)$$

- Matrix V is an $n \times n$ matrix
- Because of the orthogonality of the vectors v_i

$$V^T V = \begin{pmatrix} v_1^T v_1 & v_1^T v_2 & \dots & v_1^T v_n \\ v_2^T v_1 & v_2^T v_2 & \dots & v_2^T v_n \\ \vdots & \vdots & \ddots & \vdots \\ v_n^T v_1 & v_n^T v_2 & \dots & v_n^T v_n \end{pmatrix} = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix} = I$$

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- Eigenvectors**
- Orthogonal Matrices**
- Eigen Decomposition**
- Low Rank Approximation**



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Symmetric Matrices

- If M is an $n \times n$ **symmetric** matrix then it has n real orthogonal eigenvectors with real eigenvalues
- We denote the i^{th} eigenvector by v_i and the corresponding eigenvalue by λ_i so that

$$Mv_i = \lambda_i v_i$$

- Orthogonal means that if $i \neq j$ then

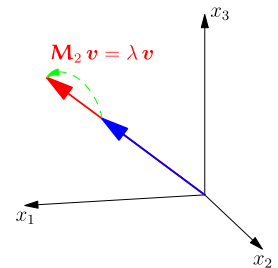
$$v_i^T v_j = 0$$

- (We can always normalise eigenvectors if we want)

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Outline

- Eigenvectors**
- Orthogonal Matrices**
- Eigen Decomposition**
- Low Rank Approximation**



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The Other Way Around

- We have shown that $V^T V = I$
- Thus multiply both sides on the left by V

$$V V^T V = V I$$

- V will have an inverse, V^{-1} , such that $V V^{-1} = I$
- Multiplying the equation on the right by V^{-1}

$$\begin{aligned} (V V^T) V V^{-1} &= V V^{-1} I \\ V V^T &= I \end{aligned}$$

- Note that, $V^{-1} = V^T$ (definition of orthogonal matrix)

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Invertible Matrices

- A matrix, M , will be singular (uninvertible) if there exists a vector $x \neq 0$ such that

$$Mx = 0$$

- Now if there exists such a vector such that $Vx = 0$ then multiply by V^T we get

$$V^T Vx = V^T 0$$

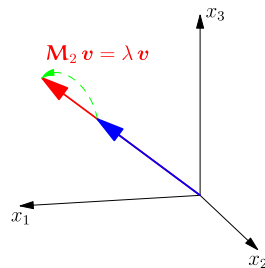
$$x = 0$$

since $V^T V = I$

- Thus V is invertible

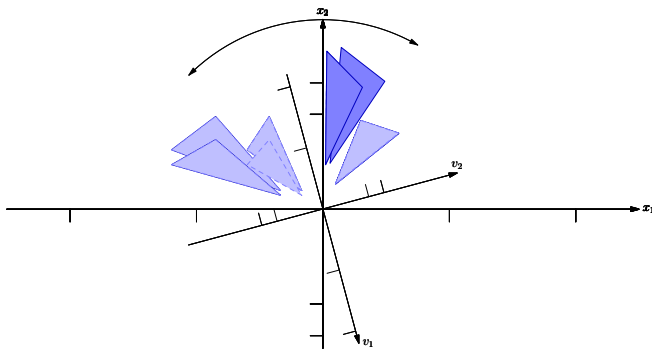
Outline

- Eigenvectors
- Orthogonal Matrices
- Eigen Decomposition**
- Low Rank Approximation



Mappings by Symmetric Matrices

$$M = \begin{pmatrix} 0.83 & -0.31 \\ -0.31 & 1.9 \end{pmatrix} = V\Lambda V^T = \begin{pmatrix} \cos(-75^\circ) & \sin(-75^\circ) \\ -\sin(-75^\circ) & \cos(-75^\circ) \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 0.75 \end{pmatrix} \begin{pmatrix} \cos(75^\circ) & \sin(75^\circ) \\ -\sin(75^\circ) & \cos(75^\circ) \end{pmatrix}$$



III-Conditioning Again

$$M = \begin{pmatrix} 0.80 & 0.01 & 0.25 \\ 0.01 & 0.85 & 0.48 \\ 0.25 & 0.48 & 0.5 \end{pmatrix} = V \begin{pmatrix} 0.8 & 0 & 0 \\ 0 & 1.25 & 0 \\ 0 & 0 & 0.1 \end{pmatrix} V^T$$

$$M^{-1} = \begin{pmatrix} 1.9 & 1.17 & -2.1 \\ 1.17 & 3.34 & -3.8 \\ -2.1 & -3.8 & 6.8 \end{pmatrix} = V \begin{pmatrix} 1.25 & 0 & 0 \\ 0 & 0.8 & 0 \\ 0 & 0 & 10 \end{pmatrix} V^T$$

$x_1 = \begin{pmatrix} -0.52 \\ -0.42 \\ 0.69 \end{pmatrix}$
 $x_2 = \begin{pmatrix} -0.7 \\ 0.17 \\ -0.61 \end{pmatrix}$
 $x_3 = \begin{pmatrix} 0.62 \\ -0.03 \\ -0.03 \end{pmatrix}$

Rotations

- Orthogonal matrices satisfy $V^T V = V V^T = I$
- As a consequent they define rotations (and possibly a reflection)
- Consider a vector x and $x' = Vx$, now

$$\|x'\|_2^2 = x'^T x' = (Vx)^T (Vx) = x^T V^T V x = x^T x = \|x\|_2^2$$

- Similarly if additionally $y' = Vy$ then

$$\langle x', y' \rangle = (Vx)^T (Vy) = x^T V^T V y = x^T y = \langle x, y \rangle = \|x\|_2 \|y\|_2 \cos(\theta)$$

- Rotations and reflections preserve lengths and angles

Matrix Decomposition

- Taking the matrix of eigenvectors, V , then

$$MV = M(v_1, v_2, \dots, v_n) = (\lambda_1 v_1, \lambda_2 v_2, \dots, \lambda_n v_n) = V\Lambda$$

- where $\Lambda = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n) = \begin{pmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_n \end{pmatrix}$

- Now

$$M = MVV^T = V\Lambda V^T$$

- Very important *similarity transform*

Inverses

- For any symmetric invertible matrix

$$M = V\Lambda V^T \quad M^{-1} = V\Lambda^{-1} V^T$$

- Where $\Lambda^{-1} = \text{diag}(\frac{1}{\lambda_1}, \frac{1}{\lambda_2}, \dots, \frac{1}{\lambda_n}) = \begin{pmatrix} \frac{1}{\lambda_1} & 0 & \dots & 0 \\ 0 & \frac{1}{\lambda_2} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \frac{1}{\lambda_n} \end{pmatrix}$

- Since

$$MM^{-1} = (V\Lambda V^T)(V\Lambda^{-1} V^T) = V\Lambda(V^T V)\Lambda^{-1} V^T = V\Lambda\Lambda^{-1} V^T = VV^T = I$$

- I.e, Small eigenvalues become large eigenvalues and visa verse

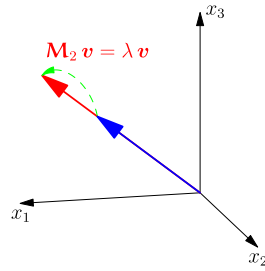
Condition Number

- Taking matrix inverses can be inherently unstable
- Any small error can be amplified by taking the inverse
- The stability of the inverse depends on the ratio of smallest eigenvalue to the largest eigenvalue (i.e. the biggest possible amplification compared to the smallest)
- For invertible matrices we can take the largest eigenvalue as a norm of the matrix
- The condition number is given by

$$\|M\|_H \times \|M^{-1}\|_H = \frac{|\lambda_{\max}|}{|\lambda_{\min}|}$$

- Large condition number implies very ill-conditioned

1. Eigenvectors
2. Orthogonal Matrices
3. Eigen Decomposition
4. **Low Rank Approximation**



- The rank of a matrix, $\mathbf{M}$, is the number of non-zero eigenvalues
- The space spanned by the eigenvectors v_a, v_b , etc. with zero eigenvalue forms a **null space**
- Any vector in the null space will get projected to the zero vector

$$\mathbf{M}(av_a + bv_b + \dots) = \mathbf{0}$$

- A square matrix is said to be **rank deficient** if it has any eigenvectors with eigenvalue equal to 0
- This happens when the columns of the matrix are not linearly independent

“Inverting” Rank Deficient Matrices

- Rank deficient matrices are non-invertible (i.e. we don't know the vector x such that $\mathbf{M}x = b$) as we don't know the component of the x in the null space
- Although we don't know x we can find a vector, x , that satisfies $\mathbf{M}x = b$
- Given a symmetric $n \times n$ matrix with k non-zero eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_k$ we can construct a “pseudo inverse” $\mathbf{M}^+$ as $\mathbf{V}\mathbf{\Lambda}^+\mathbf{V}^T$ where $\mathbf{\Lambda}^+ = \text{diag}(\frac{1}{\lambda_1}, \frac{1}{\lambda_2}, \dots, \frac{1}{\lambda_k}, 0, \dots, 0)$
- This finds the vector x with no component in the null space (it is the solution with the smallest norm)
- This is different to the pseudo inverse for non-square matrices

Summary

- Linear mappings are commonly used in machine learning algorithms such as regression
- We can understand symmetric operators by looking at their eigenvectors
- Any symmetric matrix can be decomposed as $\mathbf{M} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^T$
 - ★ where $\mathbf{V}$ are orthogonal matrices whose rows are the eigenvector
 - ★ and $\mathbf{\Lambda}$ is a diagonal matrix of the eigenvalues
- This decomposition allows us to understand inverse mappings

Low Rank Approximation

- Recall that matrices with large and small eigenvectors are ill-conditions so the inverse has the potential to greatly amplify any measurement error
- One work around is to set all small eigenvalues to zero and use the pseudo inverse
- Setting small eigenvalues to zero reduces the rank of the matrix and is an example of a low rank approximation
- Low rank approximations are much used to obtain approximate models for arrays of data (we will revisit this when we look at SVD)